

Dheer Guda

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EDUCATION

University of Maryland College Park

College Park, Maryland

B.S./M.S. in Computer Science & Math, Minor in Business Analytics & Data Science

Expected Graduation, May 2028

- **Honors:** Dean's List, QUEST Honors Program, Telora Fellowship S25, Startup Shell Grant Winner
- **Related Coursework:** Data Structures & Algorithms, Operating Systems, Database Design, Machine Learning, Artificial Intelligence, Software Engineering, Probability & Statistics

EXPERIENCE

Software Development Engineer Intern, Amazon

Seattle, Washington

LLM Deployment Cost Optimization (MARS)

- Extended MARS, Amazon's internal system for finding a model's cheapest deployment configuration, beyond weight-only optimizations (e.g. quantization) by adding knowledge distillation, its first parameter-modifying technique.
- Built hard-label, soft-label, and feature-based distillation pipelines with NVIDIA Model-Opt, Axolotl, and LLaMA-Factory to fine-tune the production Qwen 3.5 model that fills missing product type attributes (PTA) on Amazon's catalog.
- Designed a pipeline-composition framework spanning Model-Opt techniques (speculative decoding, pruning, quantization, distillation), using heuristic and surrogate-model search to find the best-serving combination.

CEO, Synari

College Park, Maryland

AI-Powered Practice Management Startup for Cognitive Therapy (synari.org)

- Grew platform to \$1,000 monthly recurring revenue (MRR) within 3 months by acquiring paying therapist clients.
- Built and launched core features (billing, treatment planning, clinical notes) with Python, FastAPI, React, and Stripe, reducing documentation time for therapists by 40%. Built and deployed with Docker on Railway through Git.
- Implemented secure HIPAA-compliant AWS infra (EC2, S3, RDS) with encryption, IAM, and VPC isolation.
- Selected for Telora Fellowship and engaged with Mokhtarzada Hatchery, refining strategy and pitching to investors.

Software Engineer, University Career Center (UCC)

Remote, New Jersey

Full-Stack Software Engineering (Academic-Year Internship)

- Built and maintained Intern for a Day (ifad.umd.edu) serving 1,000+ students/cycle on FastAPI, React/TS, PostgreSQL.
- Built custom student matching algorithm integrating Google Gemini for semantic checks, halving coordinator time.
- Led a \$35K Do Good-funded upgrade adding RBAC, FERPA-aware PII controls, audit logs, and data-retention policies.
- Shipped CI/CD (GitHub Actions + Docker) and monitoring; query tuning and caching cut p50 page load ~30%.

Startup Software Engineer Intern, SPARK

Remote, New Jersey

Full Stack Software Development Internship

- Built and deployed scalable web applications using React, Python, and FastAPI, improving UX load time by 35%.
- Integrated Stripe, AWS Lambda, and ElevenLabs voice API for secure payments, cutting latency by 20%.
- Designed backend microservices and REST APIs using PostgreSQL and Docker to support concurrent user sessions.

SAR Synthetic Dataset Research Intern, Professor Triet Le

College Park, Maryland

Software/ML Research Assistant

- Developed SAR dataset in MATLAB/Python/C to reduce data scarcity and enable self-supervised learning.
- Implemented a preprocessing pipeline (calibration, speckle filtering, normalization, tiling, augmentation) to improve model signal-to-noise and training stability.
- Benchmarked CNN baselines against classical ML (SVM/Random Forest) with k-fold CV on SAR classification, segmentation, and backscatter-regression tasks.

PROJECTS

auto-trader

College Park, Maryland

Automated Swing-Trading System

- Built a deterministic Python signals engine (momentum / RSI(2) / SPY regime filter) with inverse-volatility weighting, ATR trailing stops, and ring-fenced stock + options sleeves.
- Designed a reviewer/executor agent workflow that proposes orders as JSON, applies hard validation, and executes PASS trades via Robinhood Agentic MCP with journaled fills.

Audit.AI (Bitcamp 2024 Hackathon Winner)

College Park, Maryland

Full Stack Developer

- Built a YOLOv6-based multimodal ML system using Python and PyTorch to detect misinformation at scale.
- Engineered scalable backend with MongoDB and Redis caching for real-time media auditing, earning best overall hack.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C, SQL, R, HTML, CSS

Frameworks & Libraries: React.js, Node.js, FastAPI, Flask, React Native, PyTorch, TensorFlow, Bootstrap

Cloud & DevOps: AWS, Azure, Google Cloud, Docker, Kubernetes, Cloudflare, GitHub Actions, CI/CD, Linux, Ubuntu

Databases & Tools: PostgreSQL, MongoDB, Redis, MySQL, Jupyter, Colab, VS Code, IntelliJ, Git, Eclipse, Bash